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MTH621 QUIZ FILE

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Final project (MTH620)

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Quiz No.1, Midterm Quiz, Final Term Quiz

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Quiz No.1

If the sequence is convergent then

It is bounded

The series $\sum_{n=0}^{\infty} r^n$, -1<r<1, is

Converges and converges to $\frac{1}{1-r}$

If $f(x) = \sqrt{4-x^2}$ and $g(x) = \sqrt{x-1}$, then domain of the quotient function f/g is

(1, 2]

If $\{s_n\}$ is -------, then $\lim_{n\to\infty} s_n = inf\ \{s_n\}$.

Nonincreasing

The value of the $\lim_{n\to\infty} \frac{1}{n^p}$ is

$$\lim_{n\to\infty} \frac{1}{n^p} = 0\ for\ p > 0$$

If $\{s_n\}$ is bounded above and does not diverge to $-\infty$, then there is a unique real number $\bar{s}$ such that, if $\epsilon > 0$, $s_n < \bar{s} + \epsilon$……..

For large n

The value of the $\lim_{n\to\infty} \frac{1}{n^p}$ is

$$\lim_{n\to\infty} \frac{1}{n^p} = 0\ for\ p > 1$$

If $\sum_{n=k}^{\infty} a_n = A$ and $\sum_{n=k}^{\infty} b_n = B$, where A and B are finite, then $\sum_{n=k}^{\infty} ca_n = \cdots$

cA

if $\{a_n\}_k^{\infty}$ is an infinite sequence of the real number, the symbol $\sum_{n=k}^{\infty} a_n$ *is*

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infinite series

which statement(s) is(are) true for the following sequence $\lim_{n\to\infty} \sqrt[n]{n}$

the sequence converges to 1

if $\{x_n\}$ is ……, then $\{x_n\}$ has a convergent subsequence.

Bounded

For ……. Series $\sum_{n=1}^{\infty}\frac{1}{n} = 0$.

Harmonic

If $\sum a_n$……, then

$\lim_{n\to\infty} a_n = 0$

For the convergent sequence $\{s_n\}$, which statement is true?

$\lim_{n\to\infty} sup\, s_n = \lim_{n\to\infty} inf\, s_n = \lim_{n\to\infty} s_n$

If $|s_n| \leq r$ for any sequence $\{s_n\}$, where r is the real number, then $\{s_n\}$ is ……

Bounded sequence

If $s_n \leq b$ for any sequence $\{s_n\}$, where b is the real number, then $\{s_n\}$ is ……

Bounded sequence

A point $\bar{x}$ if a limit point of a set iff there is a sequence $\{x_n\}$ of a point in S such that $\bar{x} \neq x\ for\ n \geq 1$, and …….

$\lim_{n\to\infty} \bar{x} = x$

The sum of the convergent series is …….

Unique

The give series $\sum_{n=1}^{\infty}\frac{1}{n}$ is

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Divergent

If |x|<1, then

$$\lim_{n\to\infty} x^n = 0$$

A divergent infinite series that does not diverge to ±∞ is said to be ….

Oscillates

if $\{a_n\}_k^\infty$ is an infinite sequence of the real number, the symbol ……

$$\sum_{n=k}^{\infty} a_n$$

If $\sum_{n=k}^{\infty} a_n = A$ and $\sum_{n=k}^{\infty} b_n = B$, where A and B are finite, then $\sum_{n=k}^{\infty}(a_n + b_n) = \cdots$

A + B

If $\{x_n\}$ unbounded below, the $\{x_n\}$ has a subsequence $\{x_n\}$ such that $\lim_{n\to\infty} x_{n_k} = \cdots$

-∞

The limit of the sequence $s_n = \frac{1}{n} + \frac{2(1+3/n)}{1+1/n}$.

2

Every Cauchy sequence has a

Convergent subsequence

Grand Quiz (Midterm)

Suppose that $a_n > 0, b_n > 0$ and ……. then $\sum b_n = \infty\ if\ \sum a_n = \infty$.

$$\frac{a_{n+1}}{a_n} \le \frac{b_{n+1}}{b_n}$$

Suppose that $a_n \ge 0, b_n > 0\ for\ n \ge k$ then $\sum a_n = \infty\ if\ \sum b_n = \infty$. And

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$\lim_{n\to\infty} inf \frac{a_n}{b_n} > 0$

A sequence $\{s_n\}$ of real number is called a …… if for every $\varepsilon > 0$, there is an integer N such that $|s_n - s_m| < \varepsilon \; if \; m, n \geq N$

Cauchy sequence

The sequence $\{(-1)^n n^3\}$ is …… sequence.

Absolutely convergent

For every real x>0 and every integer n>0, there is one and only one positive real y such that…

yn = x

the series $\sum \frac{1}{n^p}$……, p>1

converges

which one is an open set of real numbers?

$\{x \in R: 0 < |x-2| < \epsilon\} \; where \; \epsilon > 0$

Which of the following statement(s) is(are) true about the following sequence $s_0 = 1$ and $s_n = 1 - e^{-s_{n-1}}$

All of the above

The following series $\sum_{n=2}^{\infty} \frac{\sin n\theta}{n+(-1)^n}$, $\theta \neq k\pi (k = integer)$,

Is convergent by Dirichlet's test

if $\{a_n\}_k^{\infty}$ is an infinite sequence of the real number, the symbol $\sum_{n=k}^{\infty} a_n \; is$ … and a_n is the nth term of the series

infinite series

the infimum of set of integers is

$-\infty$

Identify the false statement(s)

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The set of rational numbers is complete

Which sequence is bounded?

$\{(-1)^n\}$

If $\{s_n\}$ is bounded above and does not diverge to $-\infty$, then there is a unique real number $\bar{s}$ such that, if $\epsilon > 0$, $s_n < \bar{s} - \epsilon$……..

For infinitely many n

The real number system with ∞ and -∞ adjoint is called the …

Extended real number system

If $\sum a_n$……., then $\lim_{n\to\infty} a_n = 0$

Converges

Which statement is true about the sequence $s_n = (-1)^n n$

The sequence is not bounded above or below

The sum of a rational number and a irrational number is…

always an irrational number

The supremum of the set $B = \{p \in Q : p^2 < 3\}$ is

$\sqrt{3}$

The preposition p_n represented by 1 + 2 + …… + n is equal to……

n(n+1)/2

for any two reals a and b, $|a + b| \geq$ ……

||a|-|b||

The series $\sum_{n=k}^{\infty} a_n$…, if $(-1)^n\, a_n > 0, |a_{n+1}| < |a_n|,\ and\ \lim_{n\to\infty} a_n = 0$.

Converges

The number e is defined by the infinite series and denoted by ...

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$$\sum_{n=0}^{n} \frac{1}{n}$$

The set of the term of $\{s_{n_k}\}$ is contained in the set of the term of $\{s_n\}$ implies

$\sup\{s_n\} \geq \{s_{n_k}\}$

suppose that $0 \leq a_n \leq b_n, \quad n \geq k$, then

$\sum a_n < \infty \; if \; \sum b_n < \infty$

For the hormonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ =------

0

If every point of a set S is an interior point, then----

S is open

Which one is not an initial segment of N?

{2, 4, 6}

If $\{s_n\}$ converges to -1 and $\{t_n\}$ converges to 5, then the sequence $\{s_n + t_n\}$

Will be convergent and its limit is 4

If $\{x_n\}$ unbounded above, the $\{x_n\}$ has a subsequence $\{x_n\}$ such that $\lim_{n\to\infty} x_{n_k} = \cdots$

∞

Suppose -------- then the series $\sum_{n=1}^{\infty} a_n$ converges iff the series $\sum_{k=0}^{\infty} 2^k a_{2^k}$ converges.

$a_1 \geq a_2 \geq \cdots \geq 0$

For a finite set A and f from A to B is a bijection, then which of the following is true?

B is finite and |A| = |B|

Which of the following given sets is compact subset of R?

[0, 1]

A divergent infinite series that does not diverge to -----

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$\pm\infty$

If every deleted neighborhood of x_0 contains a point of S, then-----

x_0 **is a limit point of S**

The infimum of set $B = \{p \in Q : p^2 > 2\}$ is

$\sqrt{2}$

Identify the identical sequences

$\{\frac{1}{n^2-2}\}_3^\infty$ *and* $\{\frac{1}{n}\}_1^\infty$

Which one is the true for the two sets given by $S = \{x \in R : 0 < x < 1\}$ *and*

$T = \{x \in (0,1) : x \text{ is a rational}\}$?

S is a super set of T

The statement(s) true about the real number $4\sqrt{10}$ is(are)

An irrational number

Which statement is true about the sets N and Q

N⊂Q

Which set is dense in the set of real number

All of the above

If $|s_n - L| < \varepsilon$ for $n \le N$, then the sequence $\{s_n\}$ is said to be …….

Converges to the point L

If $\{s_n\}$ is the sequence of real number, then $\lim_{n\to\infty} sups_n = s$ iff

$\lim_{n\to\infty} sups_n = \lim_{n\to\infty} infs_n = s.$

=

The union of the set of rational numbers and the set of irrational numbers is

R, the set of real number

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(In comparison test) suppose the $0 \leq a_n \leq b_n,\ n \geq k, then\ if\ \sum a_n < \infty, \sum b_n < ---$

∞

$\lim_{n\to\infty} sups_n = 0$ if

$\{s_n\}$ is not a bounded above

The series $\sum \frac{r^n}{n}$ is

Divergent for 0<r<1 by comparison test

A set is closed if and only if S contains all of its ….

Limit points

If p>0 and a, is a real then value of the $\lim_{n\to\infty} \frac{n^a}{(1+p)^n}$,

0

The value of $\lim_{x\to\infty} \left(1+\frac{1}{x}\right)^x$=……

1

The preposition p_n given by 3n + 16 > 0 is true for ……

n>=-5

which of the following statement represents the completeness aziom for a nonempty set A, of real numbers?

A is bounded above and always have a superemum

The function $g(x) = x^2$ is

Increasing on [0, ∞)

Investigate the limit of $\lim_{x\to 0^+} (\frac{|x|}{x} + x)$ =------

1

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Investigate the limit of $\lim_{x \to 0^-} \left(\frac{1}{x}\right)$

-∞

Roll's theorem says, suppose that f is continuous on the close interval [a, b] and differentiate on the open interval (a, b) and f(a)=f(b)

Then $f^{/}(c) = 0$ for some c in the open interval (a, b)

The series $\sum a_n\, b_n$ converges if $a_{n+1} \leq a_n\ for\ n - - - -k, \lim_{n \to \infty} a_n = 0$, and

$|b_k + b_{k+1} + \cdots + b_n| \leq M$, for some constant M.

$\geq$

The series $\sum a_n\, b_n$ converges if $a_{n+1} \leq a_n\ for\ n \geq k, \lim_{n \to \infty} a_n = - - - -$, and

$|b_k + b_{k+1} + \cdots + b_n| \leq M$, for some constant M.

0

The inverse of the given function $f(x) = 2x + 4,\ \ 0 \leq x \leq 2$ is

$$f^{-1}(y) = \frac{y-4}{2}$$

Investigate the limit of $\lim_{n \to \infty} \frac{sinx}{x} = - - -$

1

The inverse of the function $f(x) = x^2$, is

$$f^{-1}(y) = \sqrt{y}$$

The function $f(x) = \begin{cases} x, & 0 \leq x \leq 1 \\ 2, & 1 \leq x \leq 2 \end{cases}$ is

Is nonincreasing on I = [0, 2]

Final Term Quiz

The limit $\lim_{x \to \infty} x^2 - x$ is

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Does not exist

The $lim_{x\to 2}(3x-5)=$

1

For the function $\frac{|x|}{x}$, identify the false statement(s)

$$\lim_{x\to\infty}\frac{|x|}{x}=1$$

For the function $f(x)=xsin\frac{1}{x},\ \ x\neq 0$, which statement is true

$$\lim_{x\to 0}f(x)=0$$

For the function defined as f(x)=cx, for every $\epsilon>0$ the formal definition ensures

$$|f(x)-cx_0|<,\qquad |x-x_0|<\delta, where\ 0<\delta<\varepsilon/|c|.$$

The value of $x+\lim_{x\to 0^+}\frac{|x|}{x}=$

X + 1

The limit $\lim_{x\to 1}\sqrt{x}-1$

0

If $f(x)=logx\ \ and\ \ g(x)=\frac{1}{1-x^2},\ then\ f^0g$ is

$$(f^0g)=log\frac{1}{1-x^2}$$

The $\lim_{x\to 0}\frac{4-4cosx-2sin^2x}{x^4}$ is

$$\frac{1}{2}$$

If $\lim_{x\to 0}f(x)=10$ and $\lim_{x\to 0}g(x)=-2$ then $\lim_{x\to 0}\frac{f(x)}{g(x)}$ is

-5

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The value of the limit $\lim_{x \to 0+} xlogx$

0

The function $g(x) = x^2$ is

Increasing on [0, ∞)

The derivative of the function x^n is

nx^{n-1}

The function $h(x) = -x^3$ is

Decreasing on (-∞, ∞)

If a function is continuous on the close interval then f attains its

Extreme values in the closed interval

The mean values theorem says, suppose that f is differentiable on [a, b], $f^{/}(a) \neq f^{/}(b)$, and μ is between $f^{/}(a)$ and $f^{/}(b)$.

Then $f^{/}(c) = c$ for some c in (a, b)

Generalized mean value theorem says, if f and g are continuous on the close interval [a, b] and differentiate on the open interval (a, b), then

$[g(b) - g(a)]f^{/}(c) = [f(b) - f(a)]g^{/}(c)$ For some c in (a, b).

$\lim_{x \to 0+} xlogx$, has the following indeterminate form

(0)(∞)

If $f: I \to R$ has derivative at $c \in I$, then f is

Continuous at c

The value of $\lim_{x \to \infty} \frac{e^x}{x^2}$

∞

$\lim_{n \to \infty} inf\ s_n = -\infty$ if

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$\{s_n\}$ **is not bounded below**

If $\{s_n\}$ is bounded and nonincreasing then

$\lim_{n\to\infty} s_n = inf\{s_n\}$

S = $(2,4) \cup (5,6)$ is

Open

For the piecewise defined function $f(x) = \begin{cases} x^3 & x \le 0 \\ x^2 sin\frac{1}{x}, & x > 0 \end{cases}$, the value of $\lim_{x\to 0+} f(x)$ is

0

If $f(x) = logx \;\; and \;\; g(x) = \frac{1}{1-x^2}, \;\; then\; gof$ is

$\frac{1}{1-(logx)^2}$

The inverse of the given function $f(x) = 2x + 4, \;\; 0 \le x \le 2$, is

f-1(y)=(y-4)/2

if f is a differentiable at a local extreme point$x_0 \in D_f^0$, then

$f'(x_0) = 0$

If P and P' are the partitions of [a, b], then P' is a refinement of P

If every partition point of P is also partition point of P'

If a function f is Riemann integrable on [a, b], then it is

Lipschitz continuous on [a, b].

By putting $x = sint, in\; 1 - 2x^2 = ---$

Cos2t

If $f(x) = 1 - \frac{1}{x^2}, then \lim_{x\to\infty} f(x) = --$

1

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Upper sum of f over p

$$S(P) = \sum_{j=1}^{n} M_j(x_j - x_{j-1})$$

Identify the anti-derivative of the function $f(x) = x^2 + sinx$

$$F(x) = \frac{x^3}{3} - cosx$$

For the function $F(x) = \int_0^2 f(t)dt = \frac{x^2}{2}, 0 < x \leq 1$ identify the function such that F(x) is anti-derivative of the function

F(x) = x $0 \leq x \leq 1$

For the function $f(x) = \begin{cases} x & 0 \leq x < 1 \\ 2 & 1 \leq x \leq 2 \end{cases}$ which statement is true?

The function is nondecreasing on the interval [0, 2]

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